Hands-Free Control Mechanism — Evaluation Report

Generated 2026-07-15 16:00:41 from 34556 one-second samples across 79 sessions.

Model quality — custom-trained CNNs

Blink and voice are classifiers (accuracy = fraction of correct predictions). Gaze and head pose are regressors, so their accuracy is **PCK** — the fraction of predictions within a degree tolerance of ground truth (the standard accuracy metric for gaze/pose). Generated 2026-07-15 02:25:22.

Headline	Blink	Gaze	Head pose	Voice
Accuracy 0–100%, higher is better	**98.2%** Outstanding classification accuracy	**93.5%** Excellent PCK accuracy (within 5°)	**87.7%** Good PCK accuracy (within 15°)	**92.0%** Excellent classification accuracy

Classifiers — full metric suite

Metric	Blink	Voice
Classes what it outputs	2 — open / closed 16,979 val · MRL Eye	6 — command words 2,252 val · Speech Commands v2
Accuracy 0–100%, higher is better	**98.2%** Outstanding	**92.0%** Excellent
Balanced accuracy 0–100%, per-class mean recall	**98.2%** Outstanding	**92.1%** Excellent
Error rate 0–100%, lower is better	**1.8%** Outstanding	**8.0%** Excellent
Precision (macro) 0–100%, higher is better	**98.2%** Outstanding	**92.0%** Excellent
Precision (weighted) 0–100%, support-weighted	**98.2%** Outstanding	**92.0%** Excellent
Recall / Sensitivity (macro) 0–100%, higher is better	**98.2%** Outstanding	**92.1%** Excellent
Recall (weighted) 0–100%, support-weighted	**98.2%** Outstanding	**92.0%** Excellent
Specificity / TNR (macro) 0–100%, higher is better	**98.2%** Outstanding	**98.4%** Outstanding
F1-score (macro) 0–1, harmonic mean of precision & recall	**0.982** / 1.0 Outstanding	**0.920** / 1.0 Excellent

F1-score (weighted) 0–1, support-weighted	**0.982** / 1.0 Outstanding	**0.920** / 1.0 Excellent
F0.5-score (macro) 0–1, precision-weighted	**0.982** / 1.0 Outstanding	**0.920** / 1.0 Excellent
F2-score (macro) 0–1, recall-weighted	**0.982** / 1.0 Outstanding	**0.920** / 1.0 Excellent
Jaccard / IoU (macro) 0–1, higher is better	**0.964** / 1.0 Excellent	**0.855** / 1.0 Good
Matthews corr. (MCC) −1–1, higher is better	**0.963** / 1.0 Excellent	**0.903** / 1.0 Excellent
Cohen's kappa −1–1, chance-adjusted	**0.963** / 1.0 Excellent	**0.903** / 1.0 Excellent
False-positive rate (macro) 0–100%, lower is better	**1.8%** Outstanding	**1.6%** Outstanding
False-negative rate (macro) 0–100%, lower is better	**1.8%** Outstanding	**7.9%** Excellent
False-discovery rate (macro) 0–100%, lower is better	**1.8%** Outstanding	**8.0%** Excellent
AUC-ROC 0–1, higher is better	**0.998** / 1.0 Outstanding	—
AUC-PR 0–1, higher is better	**0.998** / 1.0 Outstanding	—
AUC-ROC (one-vs-rest macro) 0–1, higher is better	—	**0.992** / 1.0 Outstanding
Top-5 accuracy 0–100%, higher is better	—	**99.6%** Outstanding
Inference time ms per frame, CPU, lower is better	**1.70 ms**	**5.50 ms**

Regressors — error & tolerance accuracy

Metric	Gaze	Head pose
Eval set	7,789 val MPIIGaze	1,969 val AFLW2000

Accuracy within 5° 0–100% of predictions (PCK)	**93.5%** Excellent	**42.8%** Weak
Accuracy within 10° 0–100% of predictions (PCK)	**99.5%** Outstanding	**75.7%** Fair
Accuracy within 15° 0–100% of predictions (PCK)	**99.9%** Outstanding	**87.7%** Good
Mean abs error degrees, 0° is perfect	**1.56°**	**8.49°**
RMSE degrees, 0° is perfect	**2.13°**	**13.55°**
Median abs error degrees, 0° is perfect	**1.21°**	**5.10°**
P90 abs error degrees, 0° is perfect	**3.25°**	**19.52°**
Mean angular error degrees, 3D direction vector	**2.43°**	**13.78°**
R^2 (best axis) 1.0 is perfect fit	**0.934**	**0.837**
R^2 (worst axis) 1.0 is perfect fit	**0.853**	**0.506**
Correlation (best axis) 1.0 is perfect	**0.967**	**0.916**
Correlation (worst axis) 1.0 is perfect	**0.924**	**0.734**
Inference time ms per frame, CPU, lower is better	**3.55 ms**	**6.14 ms**

Blink — confusion matrix

	Predicted open	Predicted closed
Actual open	**TN 8,417**	FP 174
Actual closed	FN 137	**TP 8,251**

True Positives 8,251 · False Positives 174 · True Negatives 8,417 · False Negatives 137 (positive class = “closed”)

Voice — confusion matrix

actual \ predicted	down	go	no	stop	up	yes
down	**337**	20	12	3	4	1
go	25	**320**	18	1	3	5

no	19	15	**362**	2	3	5
stop	0	3	0	**338**	9	0
up	3	3	2	3	**337**	2
yes	1	2	10	2	5	**377**

Diagonal = correct predictions; everything else is a confusion.

Model quality — pretrained & classical providers

The classical/pretrained alternatives, scored on the same public datasets as the custom CNNs so the columns are directly comparable. Generated 2026-07-15 13:26:34.

Gaze — all providers on MPIIGaze (Original)

FaceMesh detected 2268/3000 faces on MPIIGaze's own evaluation subset (much of it captured in near-dark rooms, so detection confidence is lowered to 0.15; the recovered faces were checked against MPIIGaze's manually annotated eye corners and land within 3.6 px median on a 1280x720 image, so they are real detections rather than noise). Every provider is scored on those same images through the app's real front end. The three classical providers output a screen-space signal rather than an angle, so each gets the per-user calibration the app itself performs (gain + head-pose assist + centre for the geometric/pupil mappers, the full 8-term polynomial for iris_polynomial), fitted per participant on half their samples and scored on the held-out half. The custom CNN is fitted the same way, because the app does the same thing to it (gaze.custom.gain_x/gain_y plus set_center) and because its training labels live in MPIIGaze's normalised frame, not this camera frame. IMPORTANT: this table is a harness for comparing providers against each other on equal terms, and its custom column is NOT comparable to the 93.5% / 2.43° in the regressor table above — that number scores the CNN on MPIIGaze's pre-normalised eye patches (its training domain, and the standard protocol the literature reports), whereas this one crops live from the original image via FaceMesh exactly as the running app does. The gap between the two is a genuine train/serve difference, not a scoring error.

Metric	custom (our CNN)	iris_geometric	iris_polynomial	pupil_classical
Eval set	1,130 val MPIIGaze (Original)	1,130 val MPIIGaze (Original)	1,130 val MPIIGaze (Original)	1,130 val MPIIGaze (Original)
Accuracy within 5° 0–100% of predictions (PCK)	**28.9%** Weak	**32.0%** Weak	**32.6%** Weak	**18.4%** Weak
Accuracy within 10° 0–100% of predictions (PCK)	**73.2%** Fair	**76.6%** Fair	**76.7%** Fair	**57.2%** Poor
Accuracy within 15° 0–100% of predictions (PCK)	**92.2%** Excellent	**95.2%** Excellent	**94.7%** Excellent	**86.0%** Good
Mean abs error degrees, 0° is perfect	**5.09°**	**4.85°**	**5.07°**	**6.29°**
RMSE degrees, 0° is perfect	**6.57°**	**6.85°**	**10.67°**	**7.90°**
Median abs error degrees, 0° is perfect	**4.04°**	**3.85°**	**3.75°**	**5.34°**
P90 abs error	**10.61°**	**10.05°**	**10.12°**	**12.76°**

degrees, 0° is perfect				
Mean angular error degrees, 3D direction vector	**7.92°**	**7.59°**	**7.67°**	**9.71°**
R^2 (best axis) 1.0 is perfect fit	**0.627**	**0.706**	**-0.127**	**0.395**
R^2 (worst axis) 1.0 is perfect fit	**0.405**	**0.160**	**-0.243**	**0.250**
Correlation (best axis) 1.0 is perfect	**0.794**	**0.841**	**0.566**	**0.632**
Correlation (worst axis) 1.0 is perfect	**0.636**	**0.499**	**0.440**	**0.502**

Head pose — all providers on AFLW2000

FaceMesh detected 1749/1969 faces; classical providers are scored on those (the CNN needs no landmarks). A per-axis constant offset was removed for the classical providers, mirroring the app's neutral-pose capture, and axis conventions were aligned to the dataset's once, globally (no per-sample fitting).

Metric	custom (our CNN)	solvepnp	landmark_heuristic
Eval set	1,969 val AFLW2000	1,749 val AFLW2000	1,749 val AFLW2000
Accuracy within 5° 0–100% of predictions (PCK)	**42.8%** Weak	**53.0%** Poor	**28.5%** Weak
Accuracy within 10° 0–100% of predictions (PCK)	**75.7%** Fair	**80.2%** Good	**52.6%** Poor
Accuracy within 15° 0–100% of predictions (PCK)	**87.7%** Good	**83.1%** Good	**75.4%** Fair
Mean abs error degrees, 0° is perfect	**8.49°**	**10.51°**	**10.49°**
RMSE degrees, 0° is perfect	**13.55°**	**23.76°**	**16.11°**
Median abs error degrees, 0° is perfect	**5.10°**	**3.69°**	**5.21°**
P90 abs error degrees, 0° is perfect	**19.52°**	**21.10°**	**30.25°**
Mean angular error degrees, 3D direction vector	**13.78°**	**21.34°**	**21.89°**
R^2 (best axis) 1.0 is perfect fit	**0.837**	**0.557**	**0.768**
R^2 (worst axis) 1.0 is perfect fit	**0.506**	**-0.616**	**0.285**

Correlation (best axis) 1.0 is perfect	**0.916**	**0.769**	**0.952**
Correlation (worst axis) 1.0 is perfect	**0.734**	**0.299**	**0.651**

Voice — engines on Speech Commands v2 (six command words)

vosk runs with a limited-vocabulary grammar over the same six words (its keyword-spotting mode). Both engines are scored by the same function as the custom CNNs above. One asymmetry to note: vosk can decline to answer, and did so on 40 of 900 clips — each counts as a miss (it is scored WITH its reject option, while the CNN column is pure argmax with its 0.80 runtime confidence gate switched off, so if anything vosk is handicapped here). Per-class averages cover the six real words; accuracy counts every clip.

Metric	custom (our CNN)	vosk (pretrained ASR)
Eval set	2,252 val Speech Commands v2	900 val Speech Commands v2
Accuracy 0–100%, higher is better	**92.0%** Excellent	**94.8%** Excellent
Balanced accuracy 0–100%, per-class mean recall	**92.1%** Excellent	**94.8%** Excellent
Error rate 0–100%, lower is better	**8.0%** Excellent	**5.2%** Excellent
Precision (macro) 0–100%, higher is better	**92.0%** Excellent	**99.2%** Outstanding
Precision (weighted) 0–100%, support-weighted	**92.0%** Excellent	**99.2%** Outstanding
Recall / Sensitivity (macro) 0–100%, higher is better	**92.1%** Excellent	**94.8%** Excellent
Recall (weighted) 0–100%, support-weighted	**92.0%** Excellent	**94.8%** Excellent
Specificity / TNR (macro) 0–100%, higher is better	**98.4%** Outstanding	**99.8%** Outstanding
F1-score (macro) 0–1, harmonic mean of precision & recall	**0.920** / 1.0 Excellent	**0.969** / 1.0 Excellent
F1-score (weighted) 0–1, support-weighted	**0.920** / 1.0 Excellent	**0.969** / 1.0 Excellent
F0.5-score (macro) 0–1, precision-weighted	**0.920** / 1.0 Excellent	**0.983** / 1.0 Outstanding

F2-score (macro) 0–1, recall-weighted	**0.920** / 1.0 Excellent	**0.956** / 1.0 Excellent
Jaccard / IoU (macro) 0–1, higher is better	**0.855** / 1.0 Good	**0.941** / 1.0 Excellent
Matthews corr. (MCC) −1–1, higher is better	**0.903** / 1.0 Excellent	**0.939** / 1.0 Excellent
Cohen's kappa −1–1, chance-adjusted	**0.903** / 1.0 Excellent	**0.938** / 1.0 Excellent
False-positive rate (macro) 0–100%, lower is better	**1.6%** Outstanding	**0.2%** Outstanding
False-negative rate (macro) 0–100%, lower is better	**7.9%** Excellent	**5.2%** Excellent
False-discovery rate (macro) 0–100%, lower is better	**8.0%** Excellent	**0.8%** Outstanding
Inference time ms per frame, CPU, lower is better	**2.30 ms**	**55.40 ms**

Voice — vosk confusion matrix

actual \ predicted	down	go	no	stop	up	yes	unrecognized
down	**141**	2	1	0	0	0	6
go	0	**146**	0	0	0	0	4
no	0	0	**141**	0	0	0	9
stop	1	0	0	**143**	3	0	3
up	0	0	0	0	**140**	0	10
yes	0	0	0	0	0	**142**	8
unrecognized	0	0	0	0	0	0	**0**

Diagonal = correct predictions; everything else is a confusion.

mp_geometry (head pose) reports pose relative to a per-user neutral face cloud, so it has no absolute output to score against a dataset; see the runtime tables for its live performance.

The classical blink detectors (EAR standard/adaptive, ELA) are geometric rules over FaceMesh eye landmarks, so scoring them needs full-FACE images labelled open/closed. Every eye dataset on this machine (mrlEyes_2018_01 and both archive copies) contains cropped eye images with no surrounding face, which FaceMesh cannot process, and no full-face set here (300W-LP, AFLW2000, MPIIGaze) carries eye-state labels. They are therefore compared in the runtime tables only; a labelled face set such as CEW (Closed Eyes in the Wild) would be needed to score them against the blink CNN's MRL-Eye numbers. This gap is stated rather than filled with a proxy metric.

Runtime performance — live providers

Measured on this machine from real usage sessions. These are speed/resource numbers, not model quality — see the model-quality tables above for accuracy. Averages include sessions recorded before the 2026-07-14 performance fix (single-threaded torch, throttled hand tracking, async object detection) — the custom providers' historical CPU/FPS here are far worse than the current build; click NEW SESSION on the dashboard and use the app to measure the fixed build.

Gaze — provider comparison

Metric	iris_geometric	iris_polynomial	pupil_classical	custom
Samples seconds of live data	**246 s**	**32681 s**	**322 s**	**1043 s**
FPS frames per second, higher is better	**3.59**	**21.57**	**25.05**	**14.01**
Stage latency ms per frame for this stage, lower is better	**0.03 ms**	**0.05 ms**	**1.45 ms**	**4.87 ms**
Stage p95 latency ms, 95th percentile, lower is better	**0.05 ms**	**0.13 ms**	**2.35 ms**	**10.03 ms**
Pipeline latency ms per full frame, lower is better	**272.32 ms**	**22.55 ms**	**27.03 ms**	**78.65 ms**
CPU % of one core, lower is better	**59.21%**	**56.62%**	**101.15%**	**206.28%**
Memory MB resident, lower is better	**209.33 MB**	**160.69 MB**	**329.39 MB**	**477.31 MB**
Fixation jitter px RMS while looking at one spot, lower is better	**0.41 px**	**0.18 px**	**0 px**	**1.36 px**
Calibration error px between gaze point and target, lower is better	—	**215.06 px**	**518.17 px**	—
Calibration spread px repeatability of fixations, lower is better	—	**165.63 px**	**221.24 px**	—

Blink — provider comparison

Metric	ear_adaptive	custom	ear_standard
Samples seconds of live data	**33166 s**	**938 s**	**229 s**
FPS frames per second, higher is better	**21.54**	**13.75**	**13.24**
Stage latency	**0.09 ms**	**5.33 ms**	**3.08 ms**

ms per frame for this stage, lower is better			
Stage p95 latency ms, 95th percentile, lower is better	**0.2 ms**	**13.72 ms**	**9.4 ms**
Pipeline latency ms per full frame, lower is better	**23.5 ms**	**79.36 ms**	**109.45 ms**
CPU % of one core, lower is better	**56.62%**	**204.06%**	**203.18%**
Memory MB resident, lower is better	**160.99 MB**	**481.18 MB**	**567.25 MB**

Headpose — provider comparison

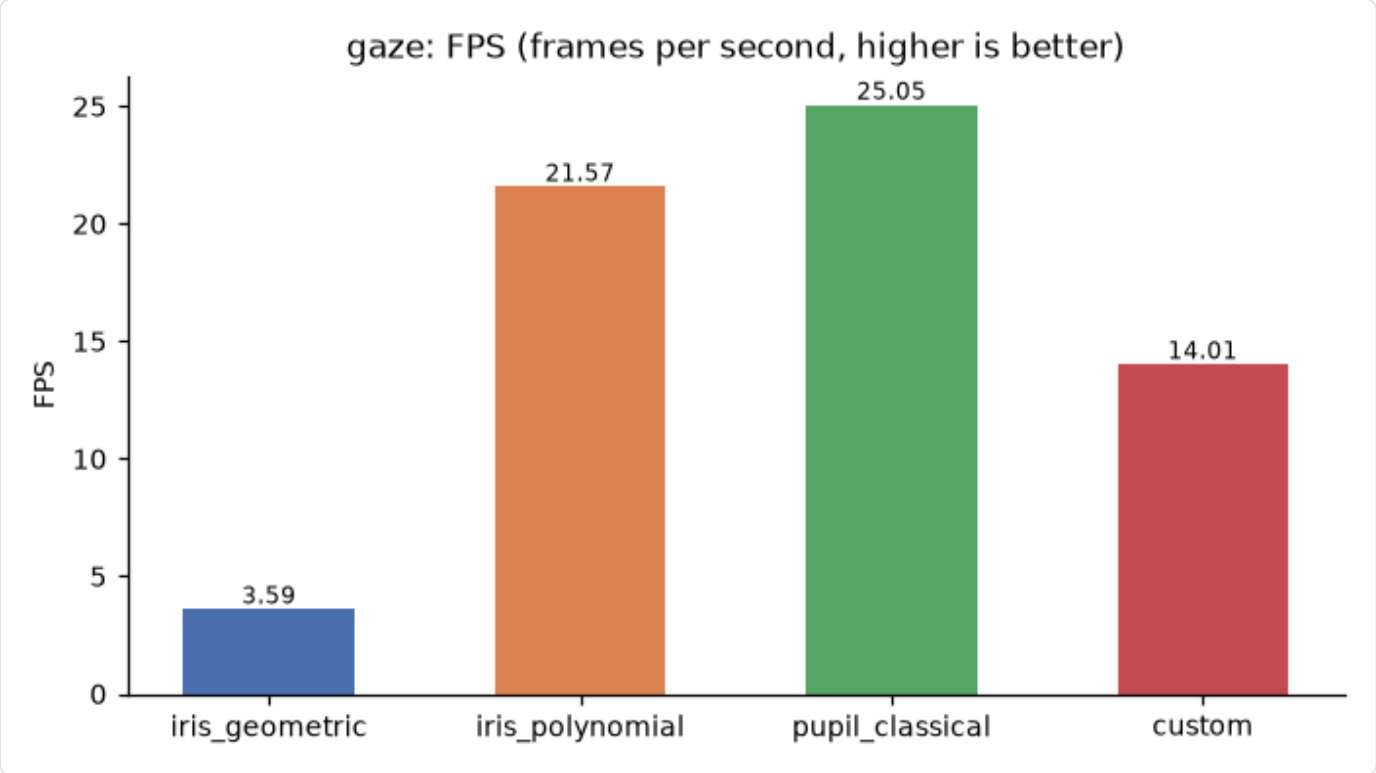
Metric	solvepnp	mp_geometry	custom
Samples seconds of live data	**9492 s**	**23475 s**	**1371 s**
FPS frames per second, higher is better	**26.72**	**19.49**	**13.54**
Stage latency ms per frame for this stage, lower is better	**0.39 ms**	**0.21 ms**	**7.38 ms**
Stage p95 latency ms, 95th percentile, lower is better	**0.91 ms**	**0.42 ms**	**17.59 ms**
Pipeline latency ms per full frame, lower is better	**11.41 ms**	**28.46 ms**	**76.25 ms**
CPU % of one core, lower is better	**49.57%**	**60.06%**	**171.86%**
Memory MB resident, lower is better	**170.55 MB**	**155.33 MB**	**479.32 MB**

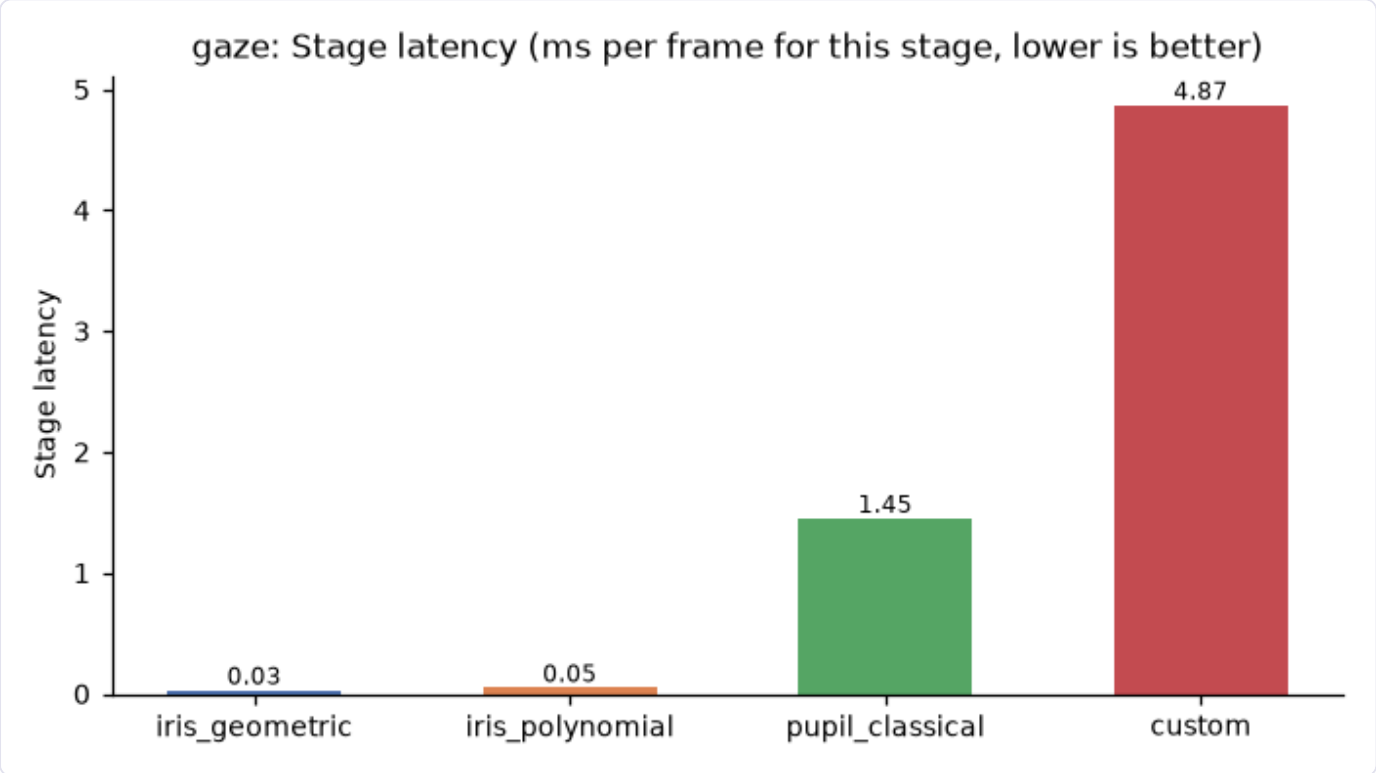
Voice — provider comparison

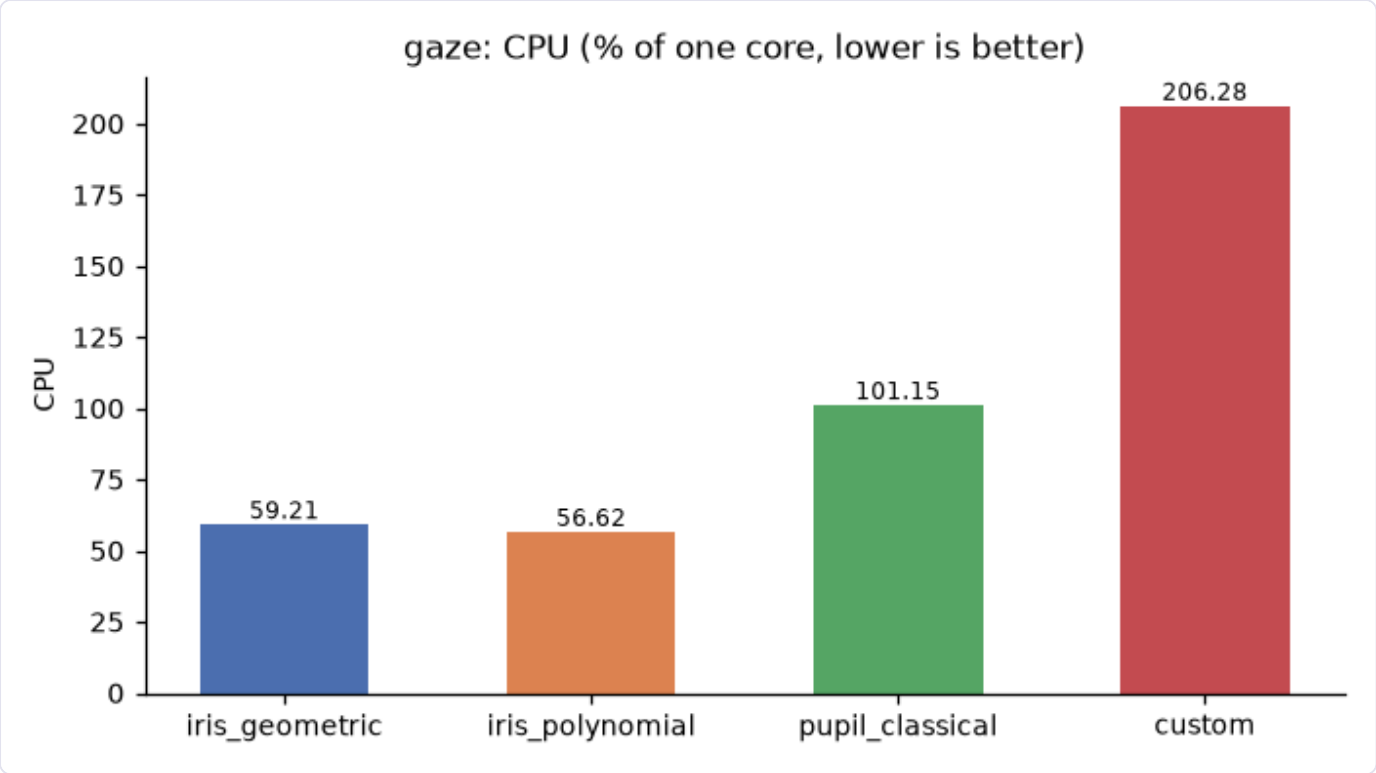
Metric	off	vosk	webspeech	whisper	custom
Samples seconds of live data	**6872 s**	**12997 s**	**13392 s**	**259 s**	**829 s**
FPS frames per second, higher is better	**18.56**	**21.71**	**22.78**	**14.61**	**13.21**
Stage latency ms per frame for this stage, lower is better	**14 ms**	**13.18 ms**	**49.36 ms**	**71.73 ms**	**9.38 ms**
Stage p95 latency ms, 95th percentile, lower is better	**82.55 ms**	**40.01 ms**	**288.96 ms**	**306.55 ms**	**27.41 ms**
Pipeline latency	**25.62 ms**	**35.3 ms**	**11.87 ms**	**46.13 ms**	**80.07 ms**

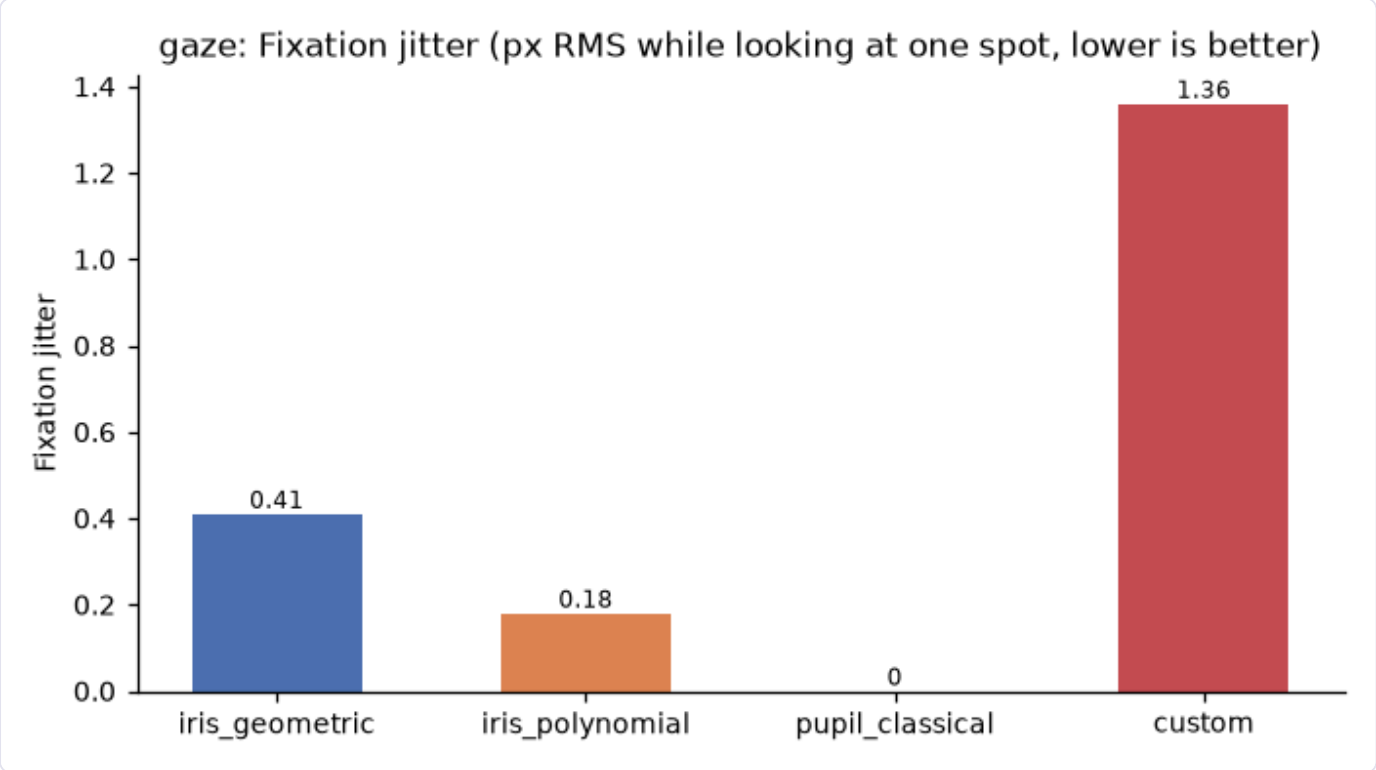
ms per full frame, lower is better					
CPU % of one core, lower is better	**53.17%**	**78.24%**	**42.8%**	**149.42%**	**150.2%**
Memory MB resident, lower is better	**124.38 MB**	**238.61 MB**	**107.73 MB**	**404.62 MB**	**505.07 MB**

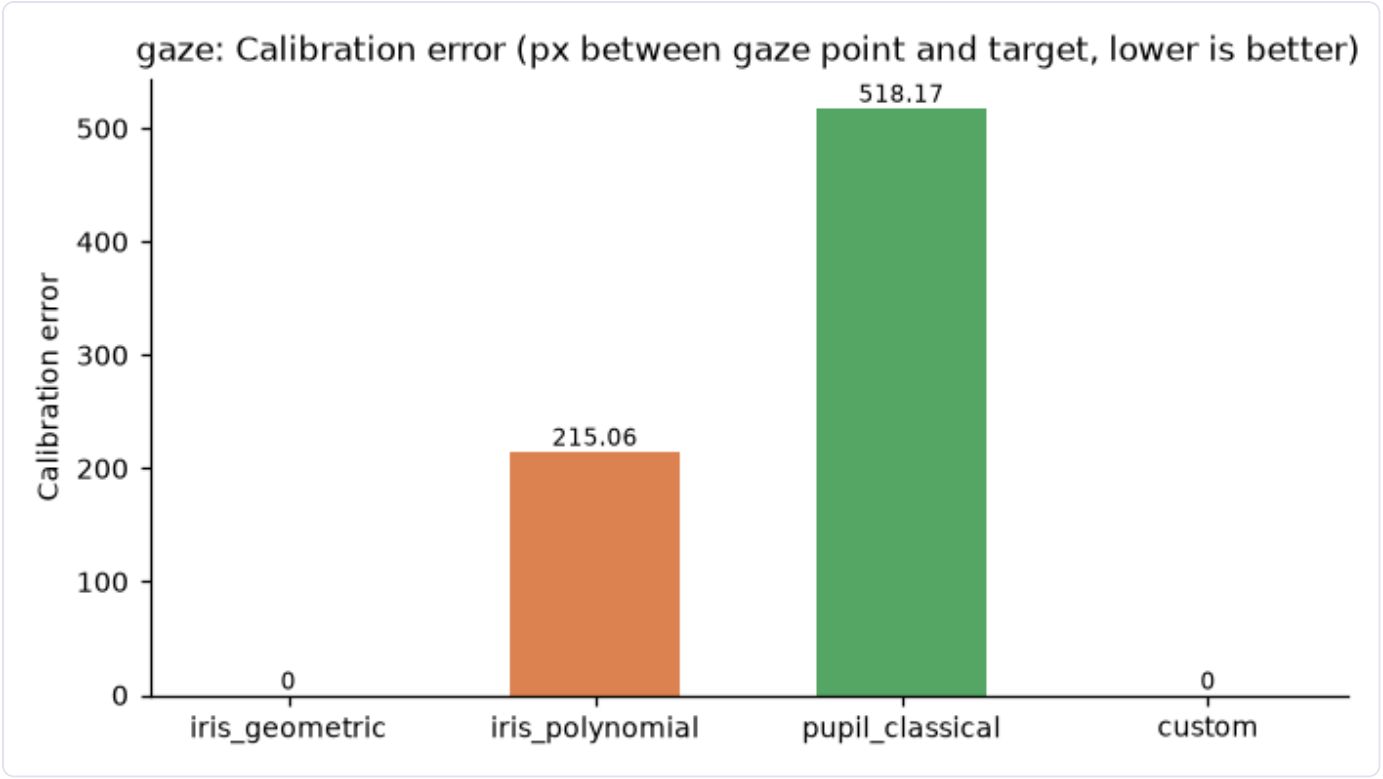
Charts

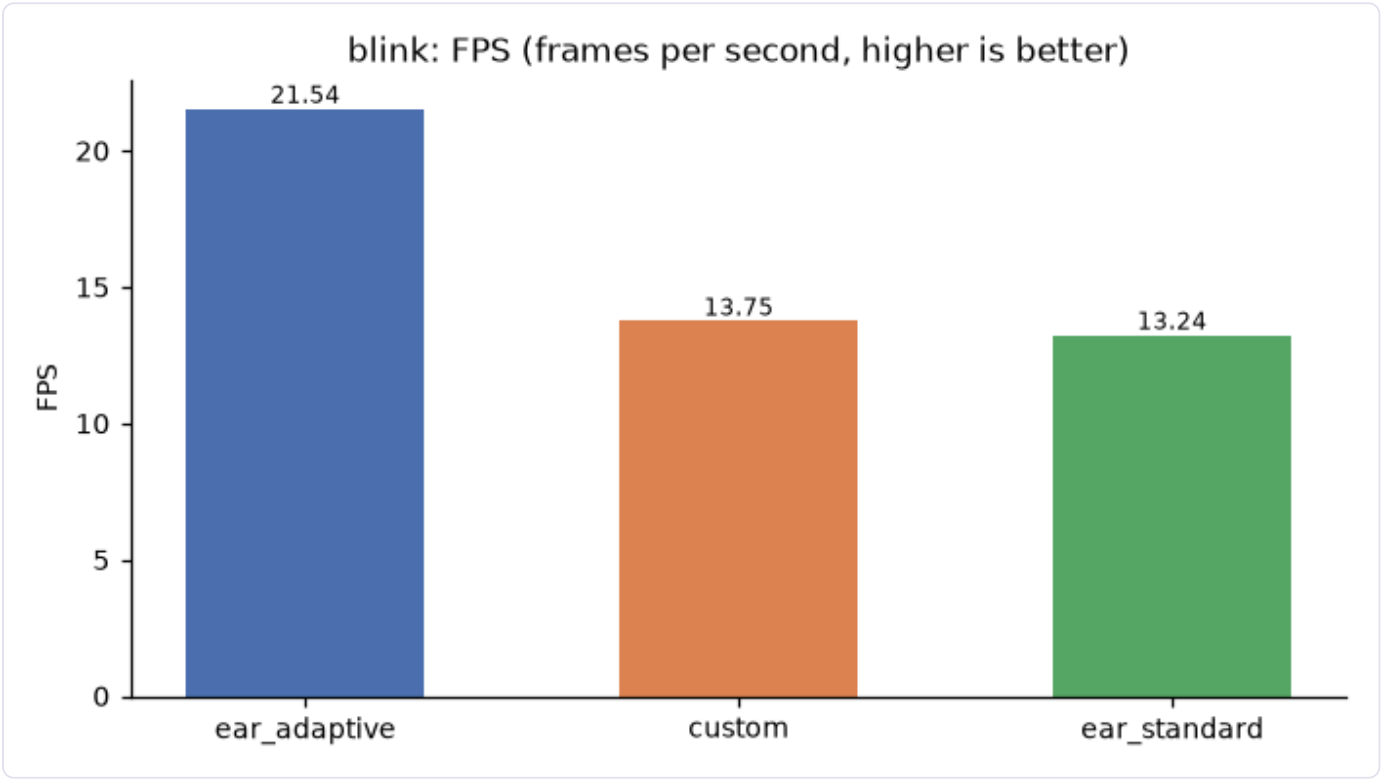


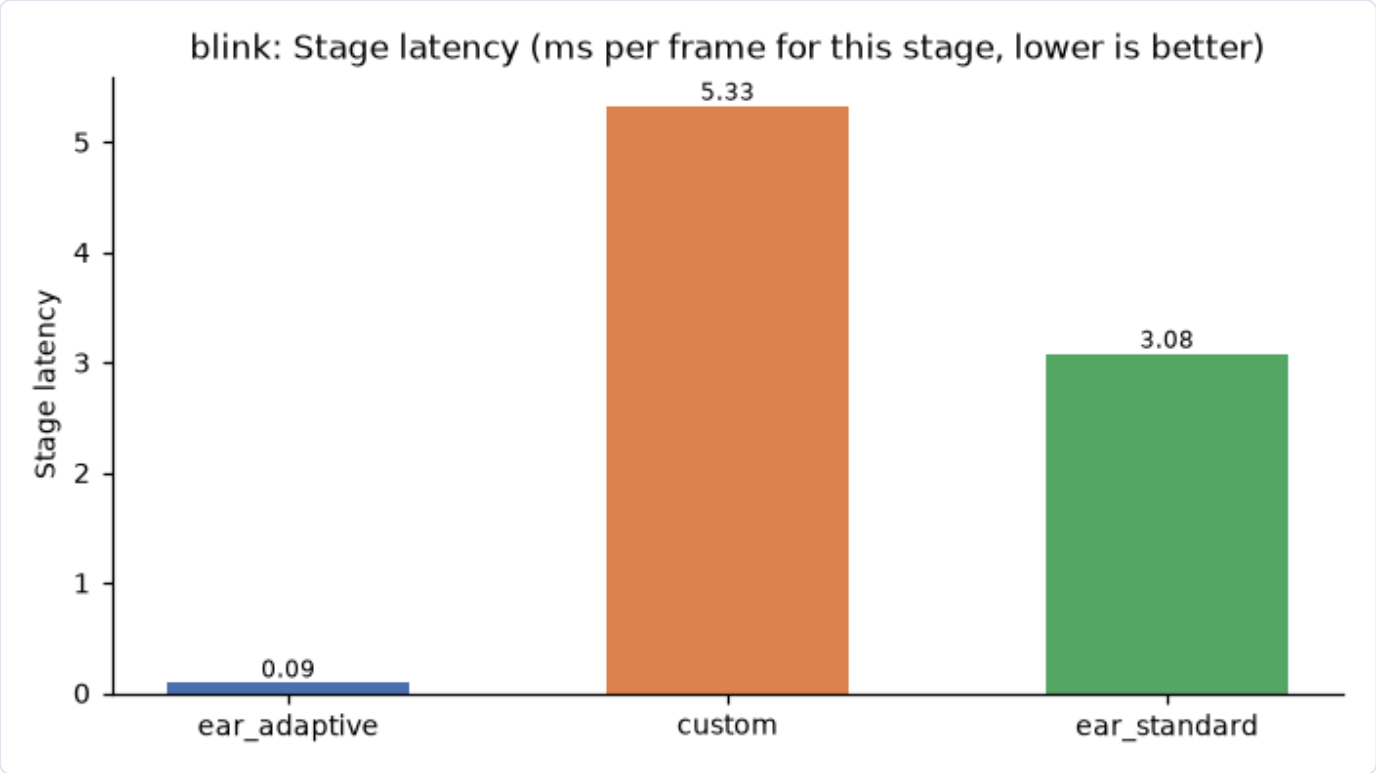


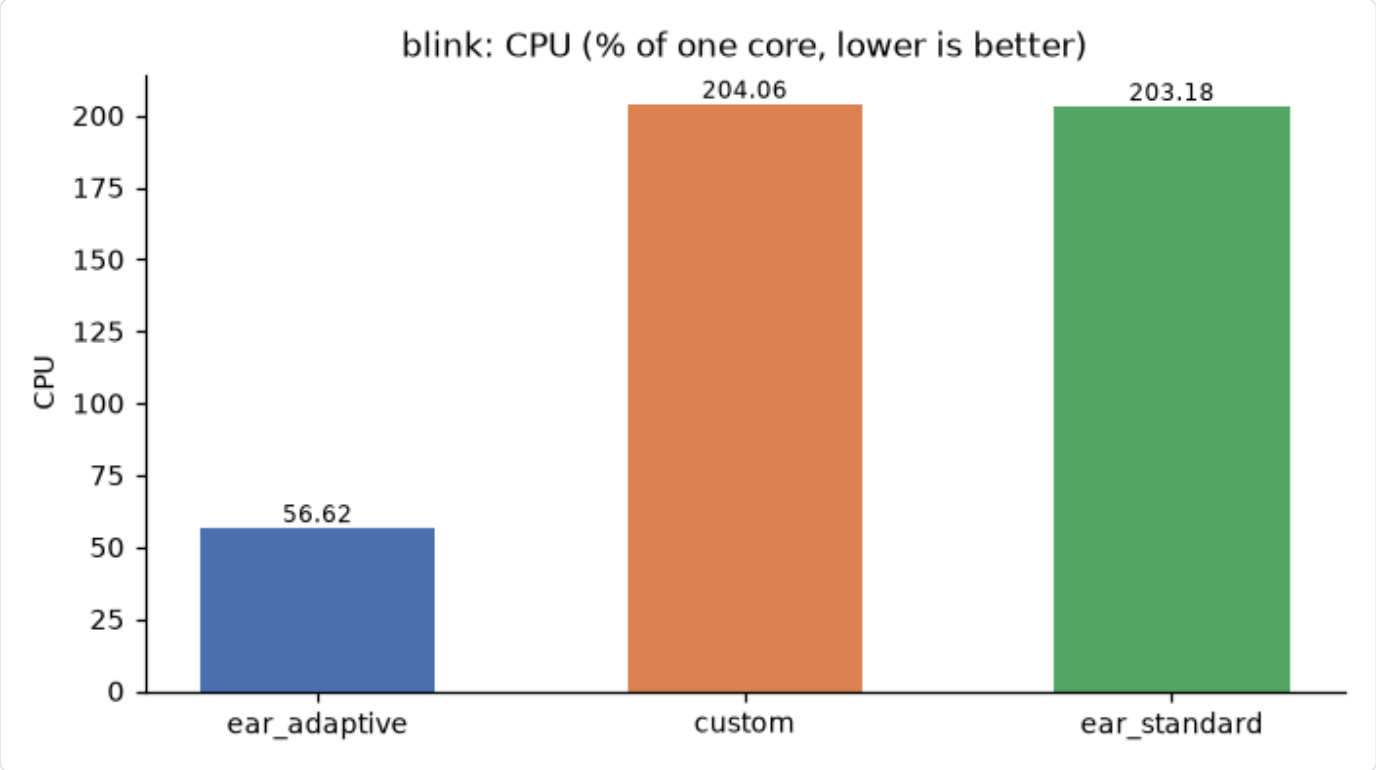

gaze: CPU (% of one core, lower is better)
CPU
0
25
50
75
100
125
150
175
200
59.21
56.62
101.15
206.28
iris_geometric
iris_polynomial
pupil_classical
custom

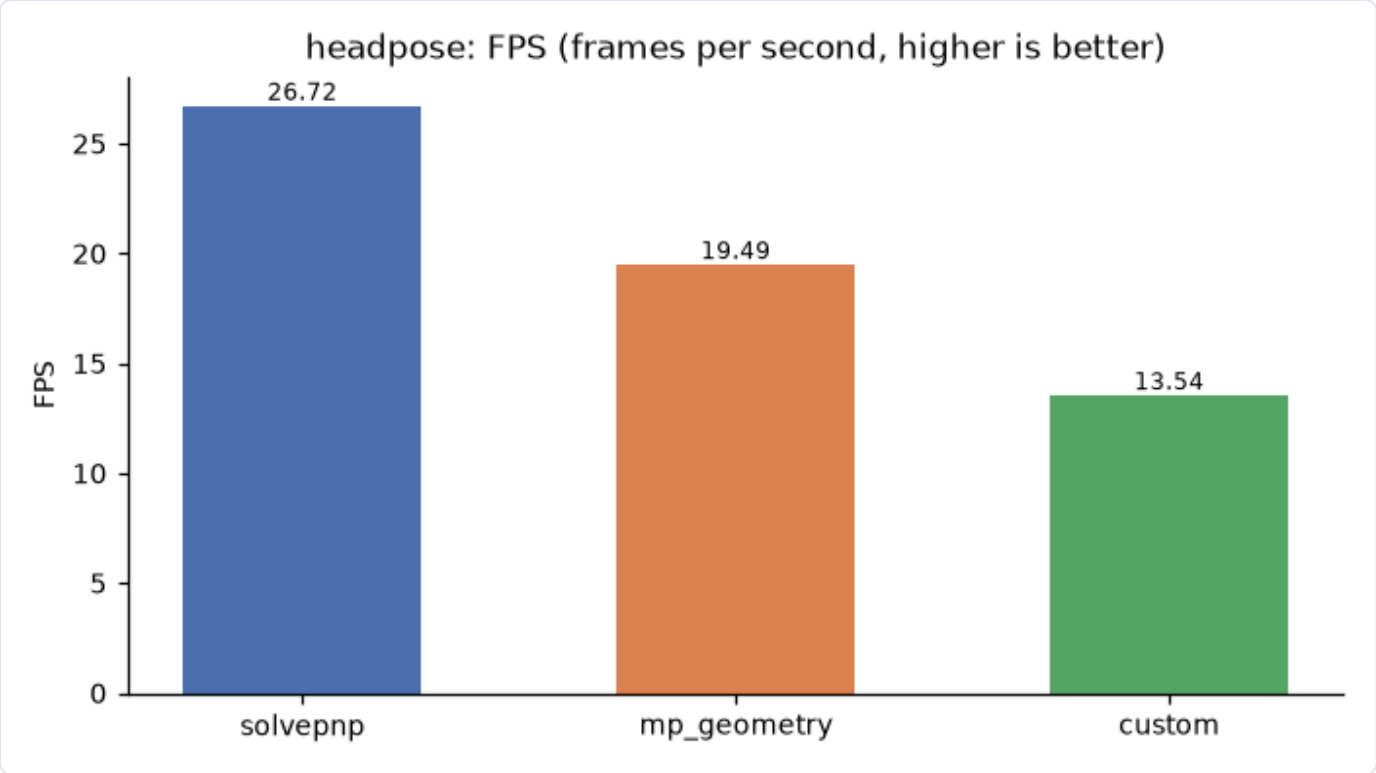

gaze: Fixation jitter (px RMS while looking at one spot, lower is better)
Fixation jitter
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0.41
0.18
0
1.36
iris_geometric
iris_polynomial
pupil_classical
custom

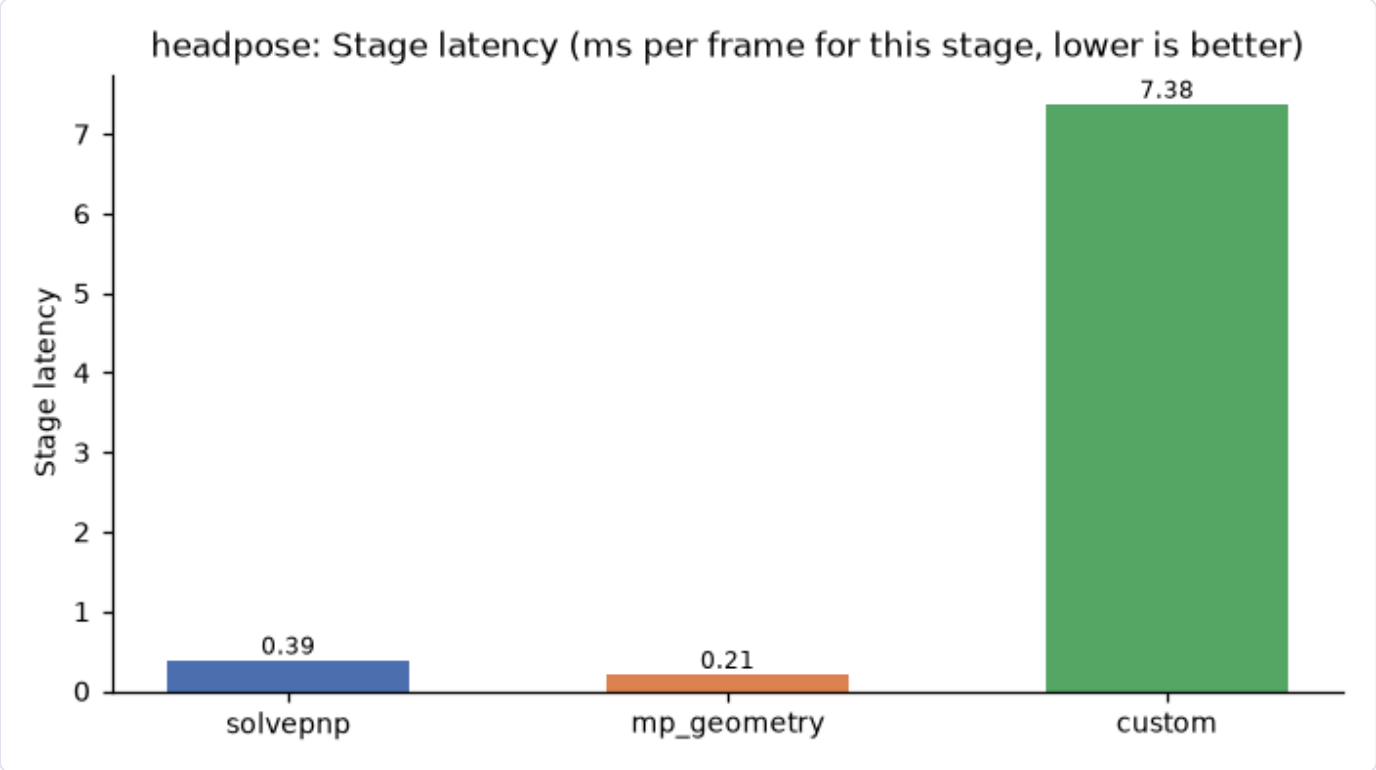

gaze: Calibration error (px between gaze point and target, lower is better)
Calibration error
0
100
200
300
400
500
0
215.06
518.17
0
iris_geometric
iris_polynomial
pupil_classical
custom

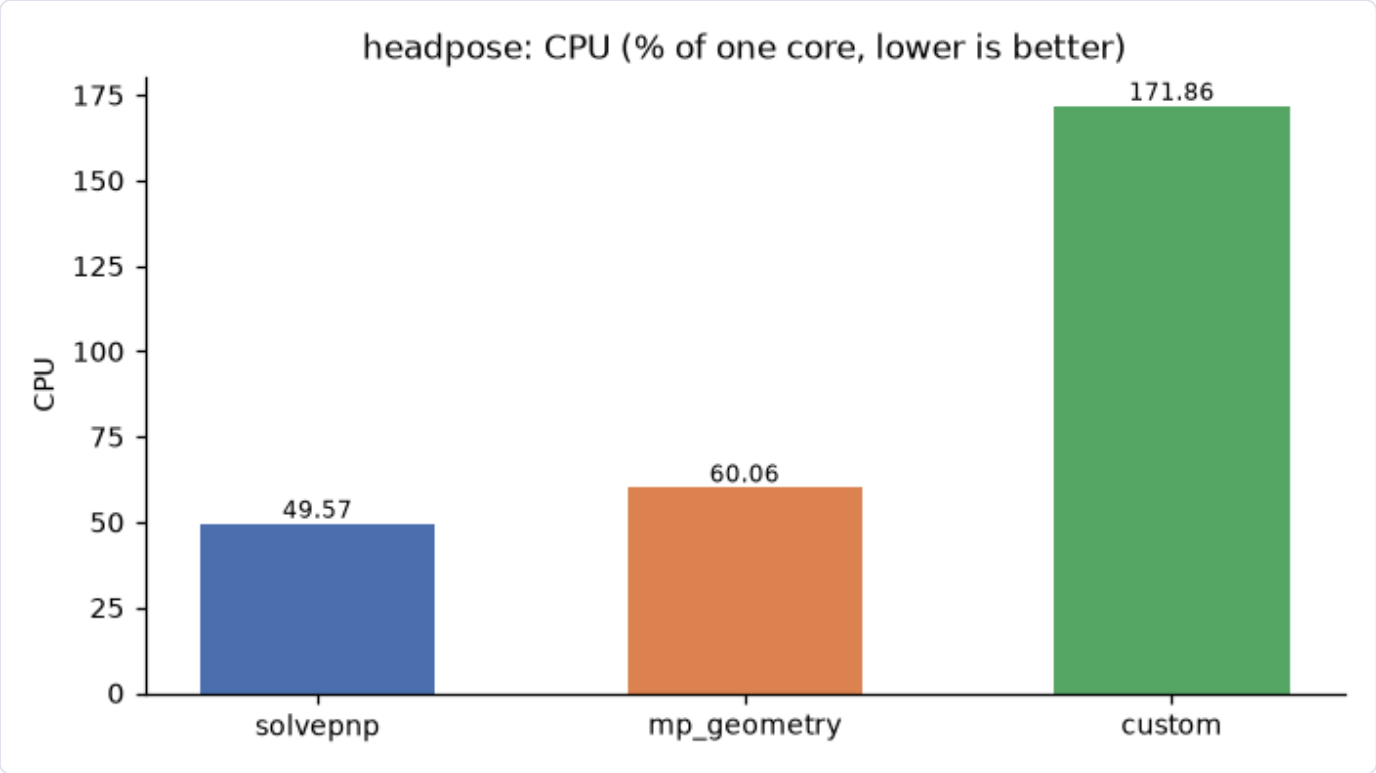

blink: FPS (frames per second, higher is better)
FPS
0
5
10
15
20
21.54
13.75
13.24
ear_adaptive
custom
ear_standard

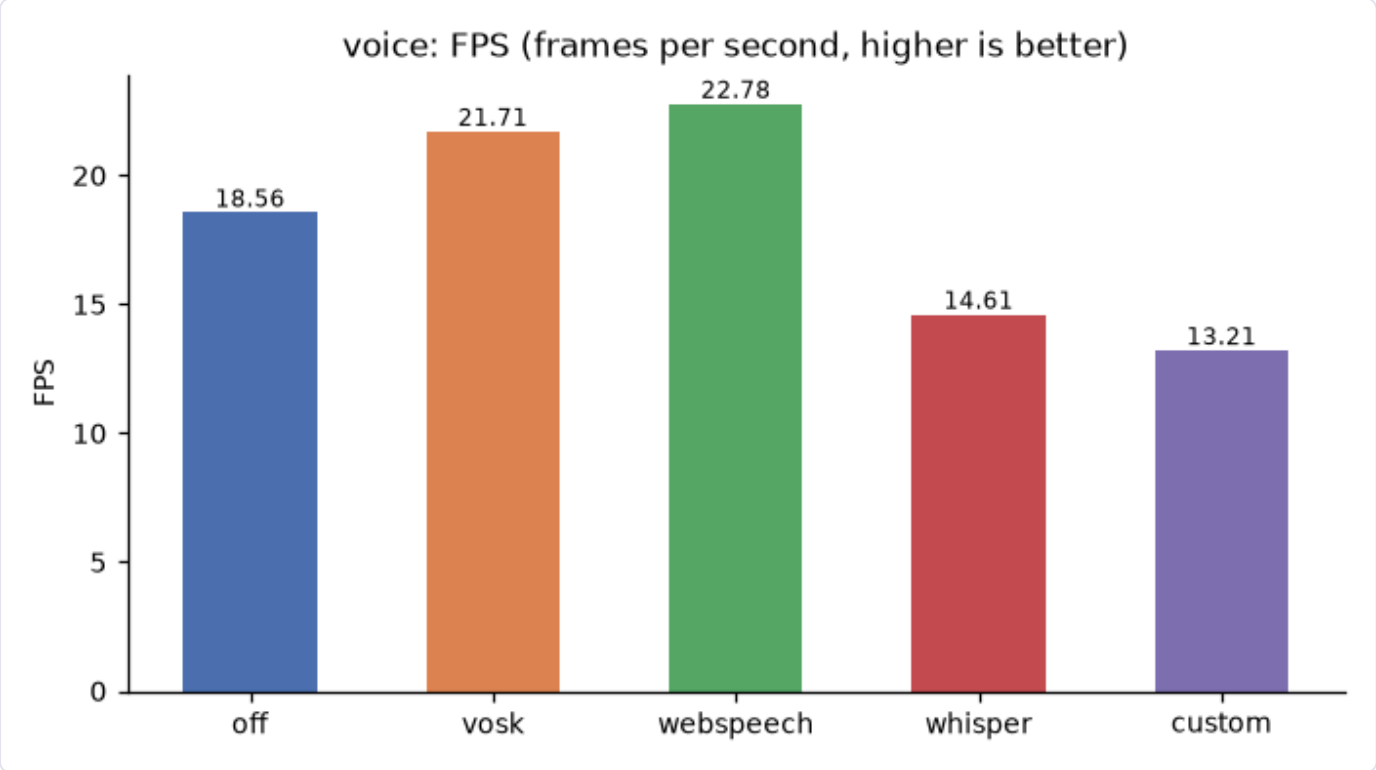

blink: Stage latency (ms per frame for this stage, lower is better)
Stage latency
0
1
2
3
4
5
0.09
5.33
3.08
ear_adaptive
custom
ear_standard

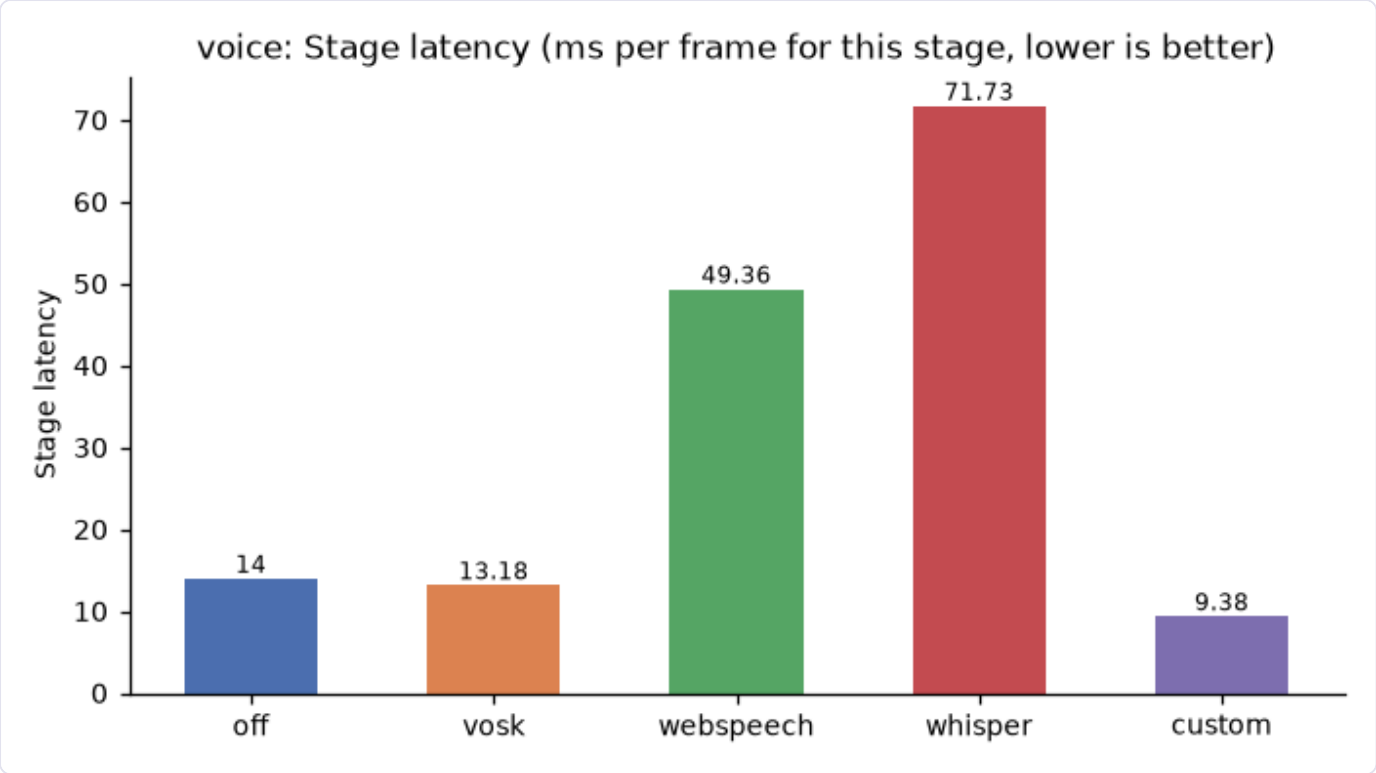

blink: CPU (% of one core, lower is better)
CPU
0
25
50
75
100
125
150
175
200
56.62
204.06
203.18
ear_adaptive
custom
ear_standard

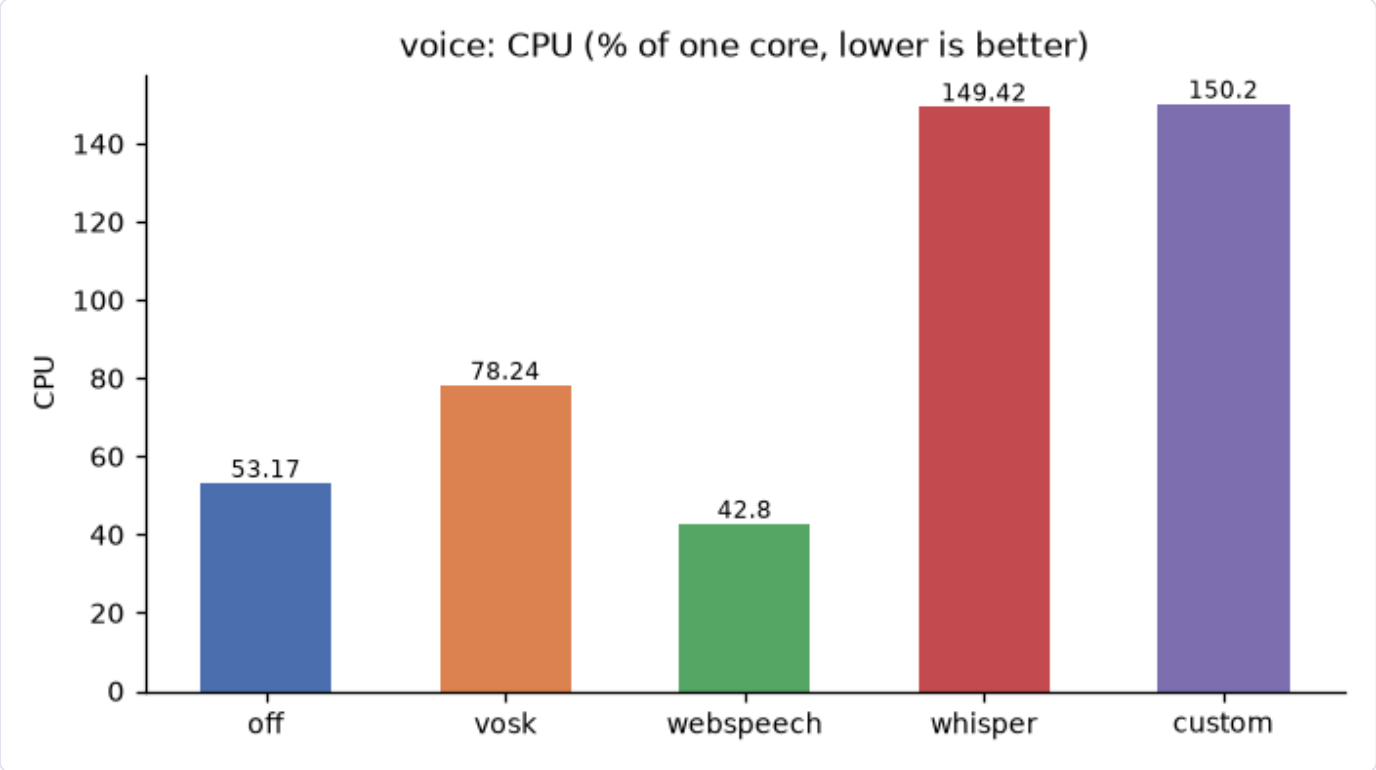

headpose: FPS (frames per second, higher is better)
FPS
26.72
19.49
13.54
0
5
10
15
20
25
solvepnp
mp_geometry
custom

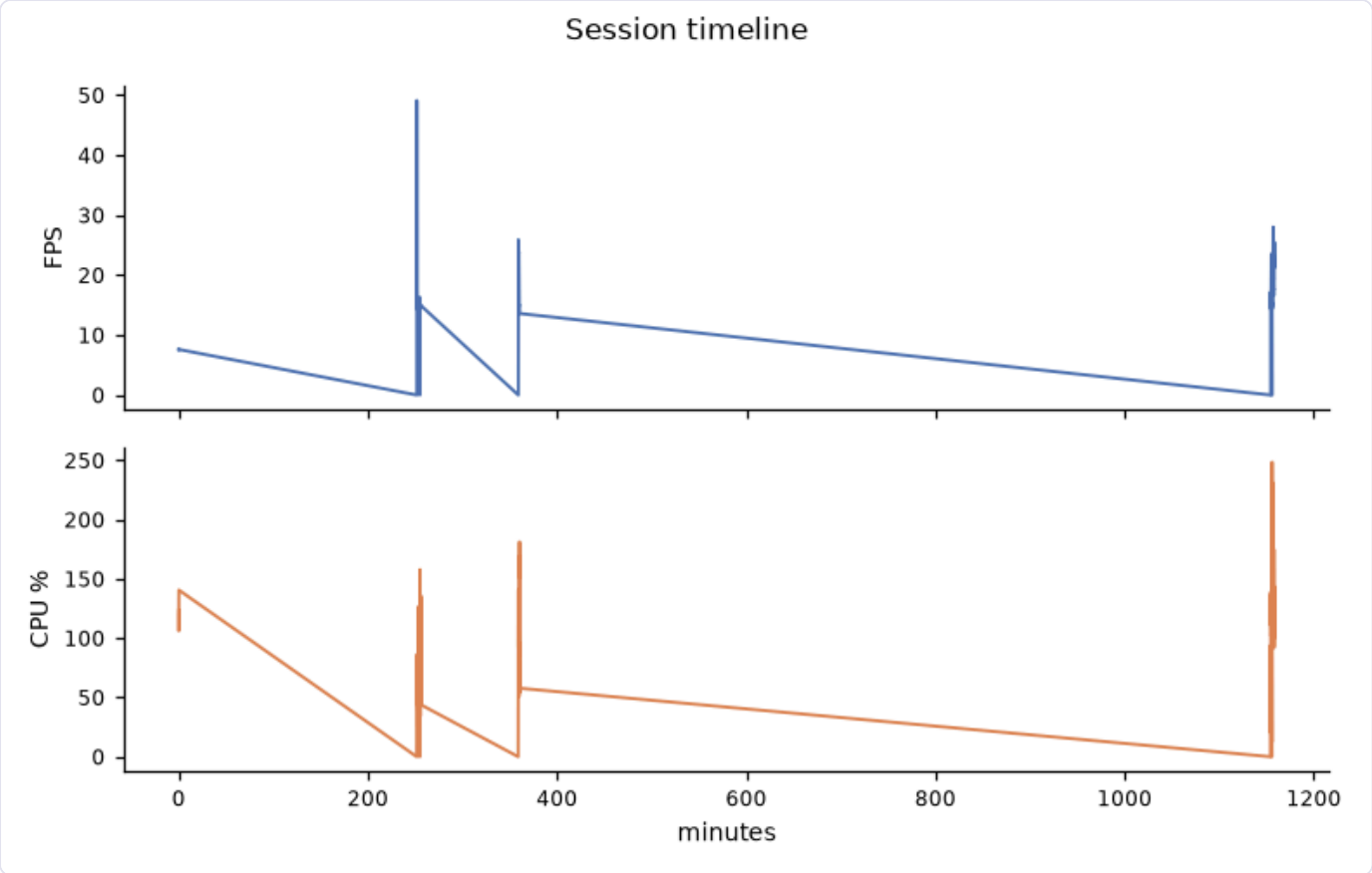

headpose: Stage latency (ms per frame for this stage, lower is better)
Stage latency
0.39
0.21
7.38
0
1
2
3
4
5
6
7
solvepnp
mp_geometry
custom


headpose: CPU (% of one core, lower is better)
CPU
175
150
125
100
75
50
25
0
49.57
60.06
171.86
solvepnp
mp_geometry
custom


voice: FPS (frames per second, higher is better)
FPS
20
15
10
5
0
18.56
21.71
22.78
14.61
13.21
off
vosk
webspeech
whisper
custom


voice: Stage latency (ms per frame for this stage, lower is better)
Stage latency
0
10
20
30
40
50
60
70
14
13.18
49.36
71.73
9.38
off
vosk
webspeech
whisper
custom


voice: CPU (% of one core, lower is better)
CPU
0
20
40
60
80
100
120
140
53.17
78.24
42.8
149.42
150.2
off
vosk
webspeech
whisper
custom


Session timeline
FPS
50
40
30
20
10
0
CPU %
250
200
150
100
50
0
0
200
400
600
800
1000
1200
minutes